

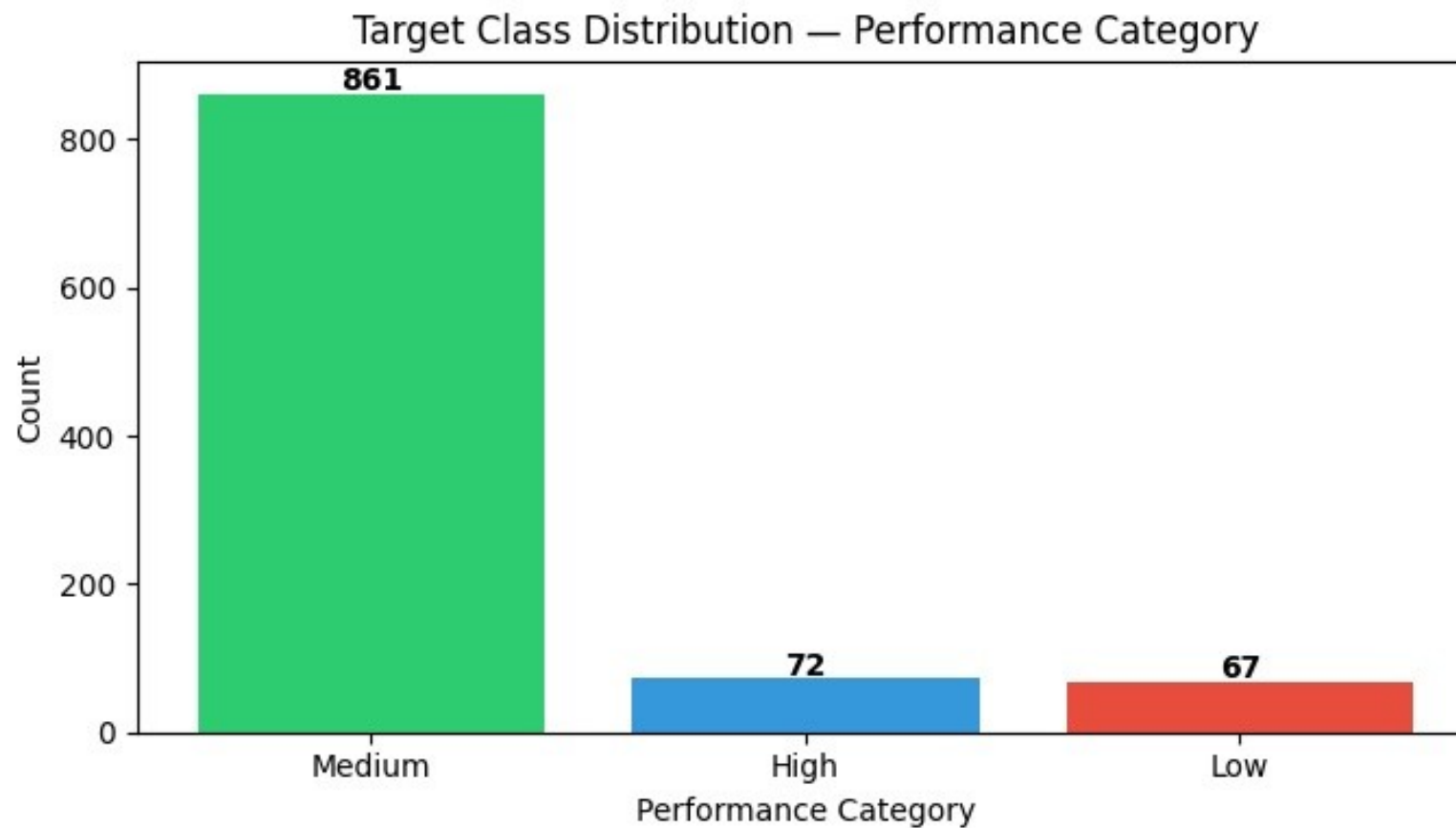
# Classification Analysis

*Categorizing Athletes: Low | Medium | High Performance*

Dataset: Kaggle — Athlete Performance Evaluation | 1,000 records | 12 features

Majd | DS301 Capstone

# Target Variable & Class Imbalance



⚠ Class Imbalance: 86.1% Medium (861 samples) vs 7.2% High (72) vs 6.7% Low (67). Handled via stratified split and weighted F1-score as primary metric — not accuracy.

# 5 Required Classification Models

## 1 Logistic Regression

Baseline, max\_iter=1000  
class\_weight='balanced'

## 2 SVC

RBF kernel, StandardScaler  
class\_weight='balanced'

## 3 Decision Tree

random\_state=42  
Visualized at max\_depth=3

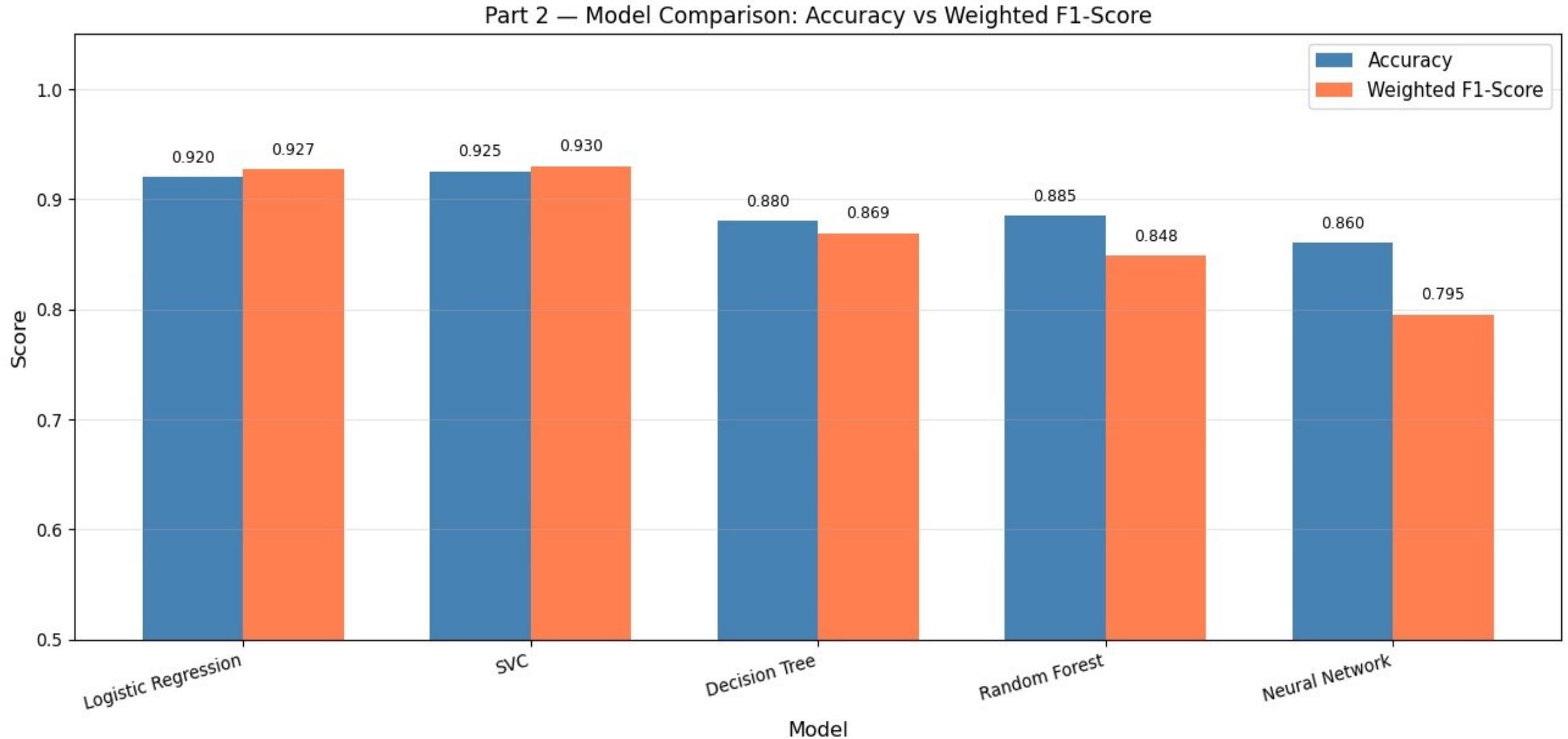
## 4 Random Forest

100 trees  
Feature importances

## 5 Neural Network

Keras: Dense(64)→Dense(32)  
→Dense(3, softmax)  
loss=sparse\_categorical\_crossentropy

# Model Comparison: Accuracy vs F1-Score



# Model Performance Results

| Model                  | Accuracy | Precision | Recall | F1-Score |
|------------------------|----------|-----------|--------|----------|
| Logistic Regression    | 0.9200   | 0.9271    | 0.9200 | 0.9271   |
| SVC (Default)          | 0.9250   | 0.9298    | 0.9250 | 0.9298   |
| SVC (Tuned) ★          | 0.9800   | 0.9803    | 0.9800 | 0.9803   |
| Decision Tree          | 0.8800   | 0.8691    | 0.8800 | 0.8691   |
| Random Forest          | 0.8850   | 0.8484    | 0.8850 | 0.8484   |
| Neural Network (Keras) | 0.9350   | 0.9316    | 0.9350 | 0.9316   |

★ Best Model: Tuned SVC (C=10, linear) — highest F1 score, primary metric due to class imbalance

# Hyperparameter Tuning — SVC

## GridSearchCV Setup

Method: GridSearchCV (5-fold CV)

Scoring: weighted F1-score

Fit on training data ONLY

### Parameter Grid:

`C = [0.1, 1, 10]`

`kernel = ['rbf', 'linear']`

`gamma = ['scale', 'auto']`

Total combinations: 12

## Best Parameters

C

10

kernel

linear

gamma

scale

## F1 Improvement

Default SVC

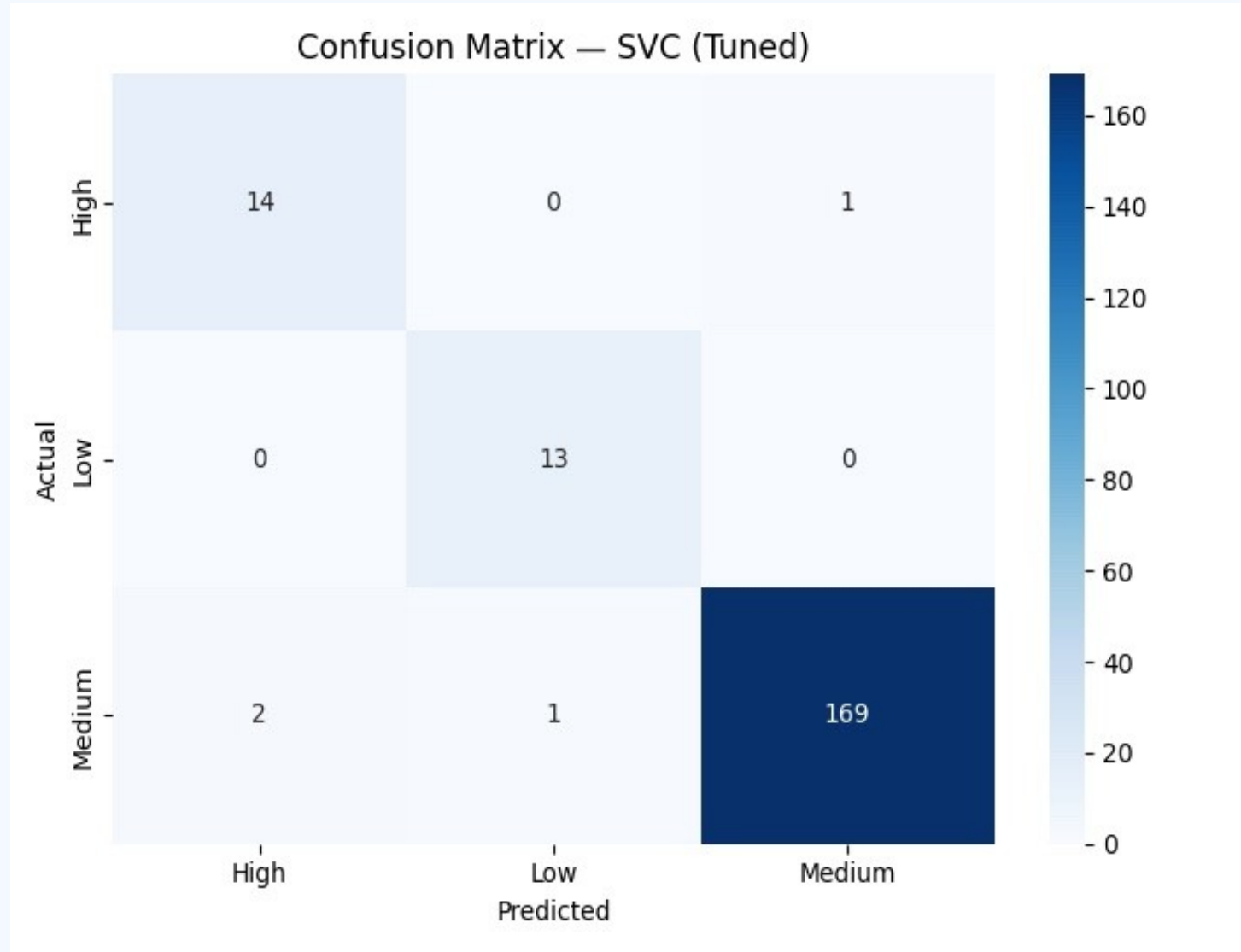
F1-Score: 0.9298

**Tuned SVC** ★

F1-Score: 0.9803

+5.3% improvement through systematic tuning

# Confusion Matrix — Tuned SVC (Best Model)



## Classification Report

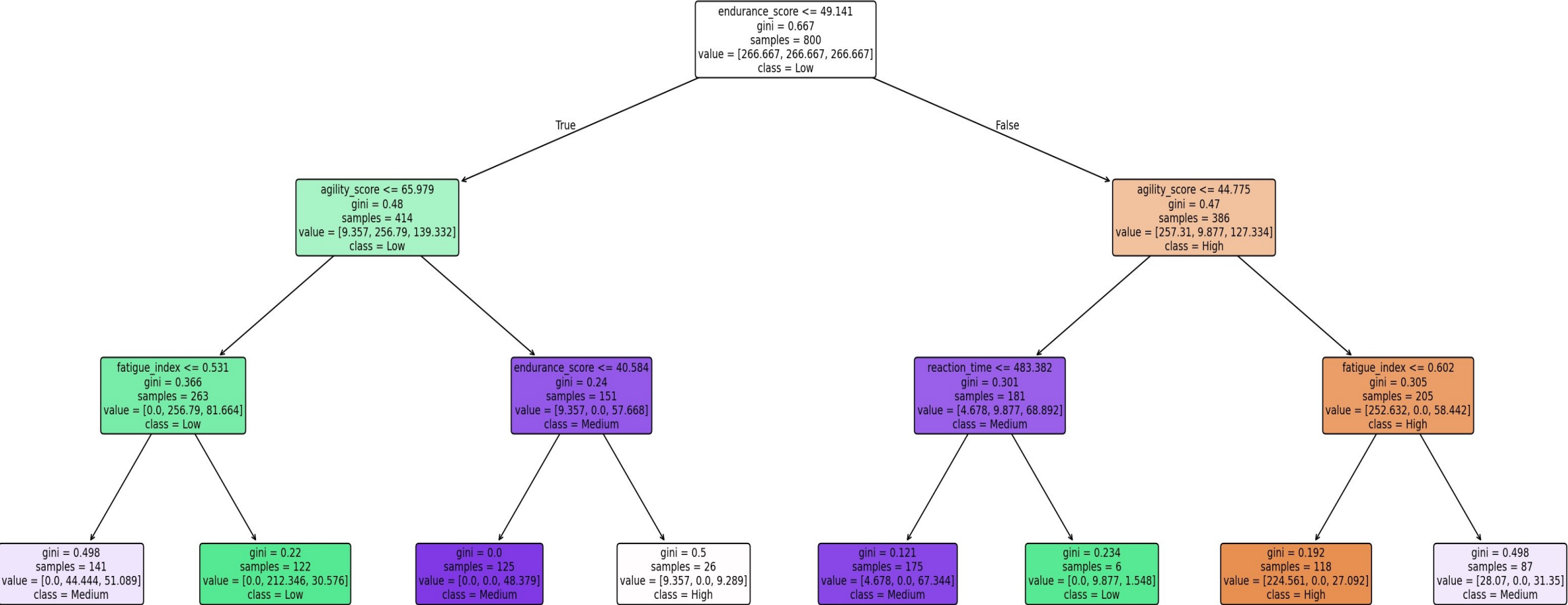
| Class   | Prec | Recall | F1   |
|---------|------|--------|------|
| High    | 0.88 | 0.93   | 0.90 |
| Low     | 0.93 | 1.00   | 0.96 |
| Medium  | 0.99 | 0.98   | 0.99 |
| Wtd Avg | 0.98 | 0.98   | 0.98 |

Overall Accuracy: 98.0%

Only 3 errors out of 200

# Decision Tree Visualization (max\_depth=3)

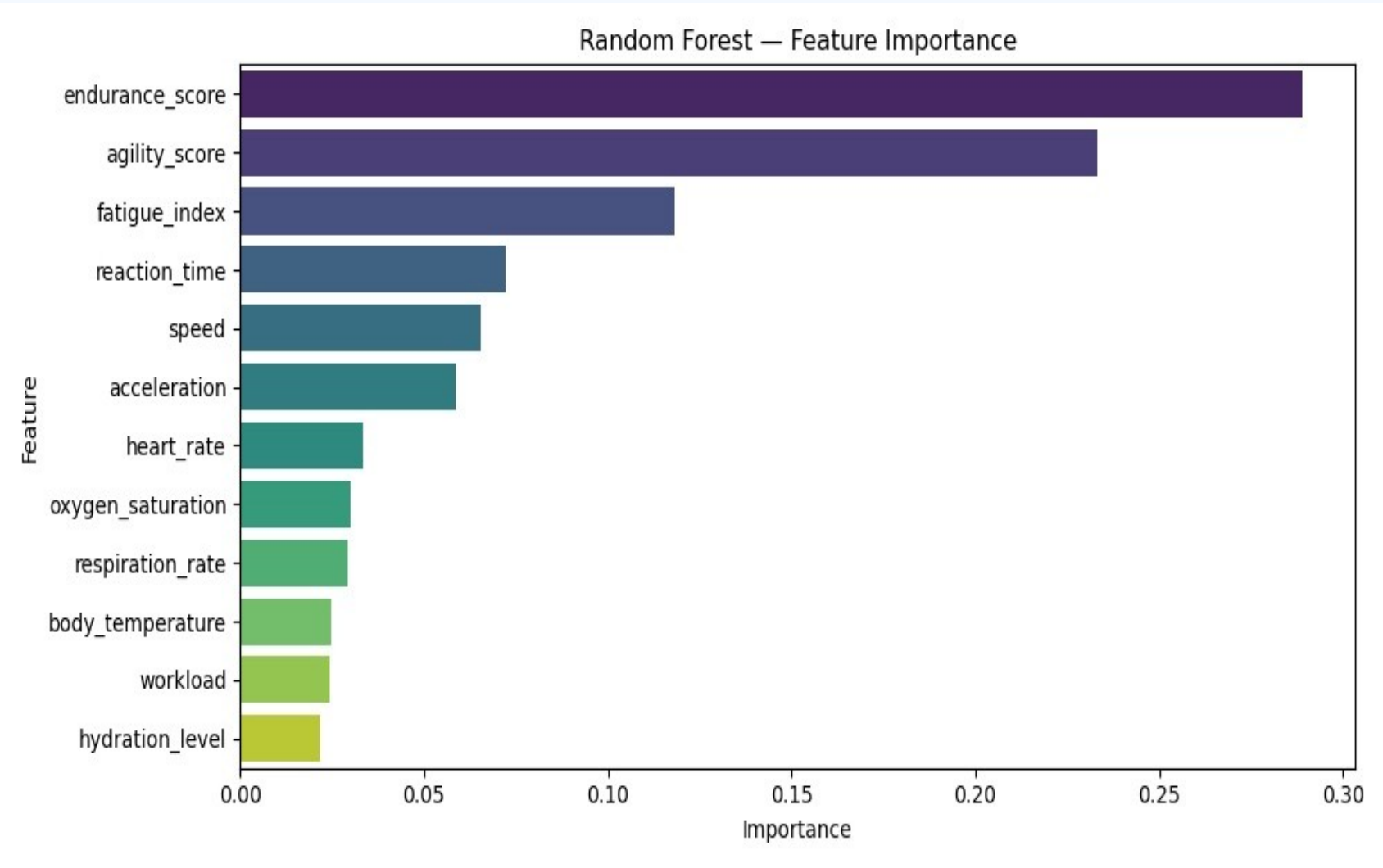
Decision Tree Visualization (max\_depth=3 for readability)



Top split: endurance\_score — confirming it is the most discriminative feature for athlete categorization.



# Feature Importance — Random Forest Classifier



## Key Insight

endurance\_score **28.9%**

agility\_score **23.3%**

fatigue\_index **11.8%**

Top 2 features  
account for >52%  
of predictive power.

Matches regression  
results — consistent  
feature ranking.

# Conclusion

## 5 Models Trained

Logistic Reg, SVC,  
Decision Tree, Random Forest,  
Neural Network (Keras)

## Best: Tuned SVC

C=10, linear kernel  
98% accuracy, F1=0.9803

## Class Imbalance

86.1% Medium handled via  
stratified split + F1 metric

## Key Features

Endurance + agility  
drive >52% of classification